

Outputs Explanation

The document goes through every output file produced by the programs in the source code archive and explains what is inside it, how to read it, and what it told us. Figures are described but not shown here; the files themselves sit in the numbered output folders.

cross_dataset_direction_check.py

Since nothing is significant at 3 months, the follow-up question is whether there is nevertheless a faint hint of the 18-month change already present in the young fish, too small to reach significance. The script tests that by taking the genes that are significant at 18 months and looking at which way they are already leaning at 3 months.

- **cross_dataset_direction_summary.csv:** per gene, its 3-month fold change, its 18-month fold change, the 18-month significance, and which 18-month group it belongs to (up in mutant, down in mutant, or not significant). This is the table behind all four plots below.
- **cross_dataset_density.png:** smoothed distributions of the 3-month fold changes, drawn separately for the genes that end up going up at 18 months, the genes that end up going down, and everything else as a background. If the up and down curves are shifted away from the background, the drift is already there in the young fish.
- **cross_dataset_boxplot.png:** the same comparison as boxplots with the group means written on, which makes the size of any shift easier to read off than the density curves.
- **cross_dataset_scatter.png:** the 18-month fold change on one axis against the 3-month fold change on the other, for the shared genes, with the correlation printed on the plot. A positive correlation would mean the young fish are already moving in the same direction as the old ones.
- **cross_dataset_cdf.png:** the cumulative version of the density plot. Curves that are separated left to right mean the groups genuinely differ, and the cumulative form makes small consistent shifts easier to see than a histogram does.